

AZARIA A. REED

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PROFESSIONAL EXPERIENCE

Systems Engineer

Jan 2022 – Present

SafeHug LLC

Madison, AL

- Owned verification and validation of a production web system supporting a patented medical safety device, ensuring alignment with real-world safety intent
- Led change management and continuous monitoring, identifying risks, anomalies, and emerging issues to preserve system stability, security, and operability
- Served as the primary technical point of contact, ensuring traceability between requirements and system behavior
- Validated end-to-end system functionality, performance, and operational reliability, translating stakeholder intent into verifiable technical outcomes

Software Development Engineer in Test (SDET)

May 2026 – Present

Octave Intelligence plc • Internship

Madison, AL

- Scaled Smart 3D verification by engineering an AI-driven generation workflow projected to reduce engineering effort across 3 feature areas (58 work items) by approximately 67%, generating production-ready C# test suites in about 1 hour per class instead of 30 developer-days
- Engineered an AI-driven generation-and-verification workflow for Smart 3D, Octave's flagship industrial engineering design platform, evolving it through 14 production versions to generate 67 passing C# unit tests with 100% line coverage while introducing modular orchestration and deterministic verification
- Diagnosed and systematically eliminated 84 compilation errors and 50+ production runtime failures by engineering deterministic verification rules for COM interoperability, external assembly loading, analyzer compliance, initialization, constructor sequencing, and static-state isolation
- Delivered Smart 3D ecosystem enhancements by validating production behavior across 37 comprehensive scenarios, refining them into 34 mutually exclusive production test cases, and identifying an additional application freeze defect

Software Engineering Contractor

May 2025 – Aug 2025

Tektonux LLC

Huntsville, AL

- Authored technical documentation for C# unit, end-to-end, and Storybook tests, enabling efficient onboarding and test adoption
- Supported automated test design and execution (Playwright) to validate defense system behavior against visual and functional requirements
- Tested terrain server functionality using GeoTIFF DEMs and Cesium, ensuring replicability across environments
- Implemented user interfaces in TypeScript, CSS, and HTML, ensuring accuracy across development stages

Test Engineer

Dec 2022

Independent Client (E-Commerce)

Dallas, TX

- Released a customer-facing production web system, validating readiness and enabling immediate deployment and revenue impact
- Partnered directly with the business owner, confirming requirements and ensuring delivered behavior matched business expectations
- Tested system behavior, usability, and visual correctness to ensure production readiness prior to launch

EDUCATION

Bachelor of Science in Computer Science

The University of Alabama in Huntsville

Huntsville, AL

- **Minor:** Mathematical Sciences | **Concentration:** Artificial Intelligence & Data Science
- **Summa Cum Laude**

Associate of Science in Computer Science

Calhoun Community College

Decatur, AL

- **Emphasis:** Business Information Technology
via CIS Certificates — Python, Java, C#, C++; MOS: Associate — Excel, PowerPoint, Word
- **Summa Cum Laude | Top Ten Student Award**

AREAS OF EXPERTISE

- **Leadership & Management:** Technical Leadership, Stakeholder Management, Change Management, Technical Communication, Cross-Functional Collaboration, Technical Documentation
- **AI & Automation:** AI-Driven Engineering, AI-Driven Test Generation, Engineering Automation, Test Automation, Deterministic Verification, Workflow Orchestration
- **Systems & Testing:** Verification & Validation (V&V), Requirements Traceability, Root Cause Analysis, Systems Testing, Test Case Design, End-to-End Testing, Integration Testing, Regression Testing, Unit Testing